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Ready to code? Let's get our hands dirty!

Download Processing from the link. On Windows, when you click on the link, a .zip file will be downloaded. Double-click it to extract the files and click *Processing.exe* to start (<https://download.processing.org/processing-3.5.4-windows64.zip>).

The Processing equivalent of a 'Hello World' program is simply to draw a line:

```
line(1,2,5,2)
```

This code calls Processing's line function, which tells the computer to draw a line.

This line of code means, 'Draw a line, from coordinate (1,2) to (5,2)'. Line of code represents a line (x1, y1, x2, y2) where (x1, y1) are coordinates of the first point, and (x2, y2) are the coordinates of the second point. Click *Run* (the triangle button in the toolbar), and your code gets compiled and executed. By nature, Processing programs never get terminated; they run forever and ever until they get disturbed. You can terminate them programmatically; however, if you don't, you can use the *Stop* button. You've used the Processing editor to write your first line of code. Congratulations!

Now, let's understand the structure of the program in Processing. Processing code consists of two main parts, *setup()* and *draw()* blocks. The *setup* block runs once when the code gets executed, and the *draw* blocks run continuously.

```
void setup() {
  size(500, 500);
  background(255, 128, 0);
  stroke(0);
}
```



Figure 1: Download Processing

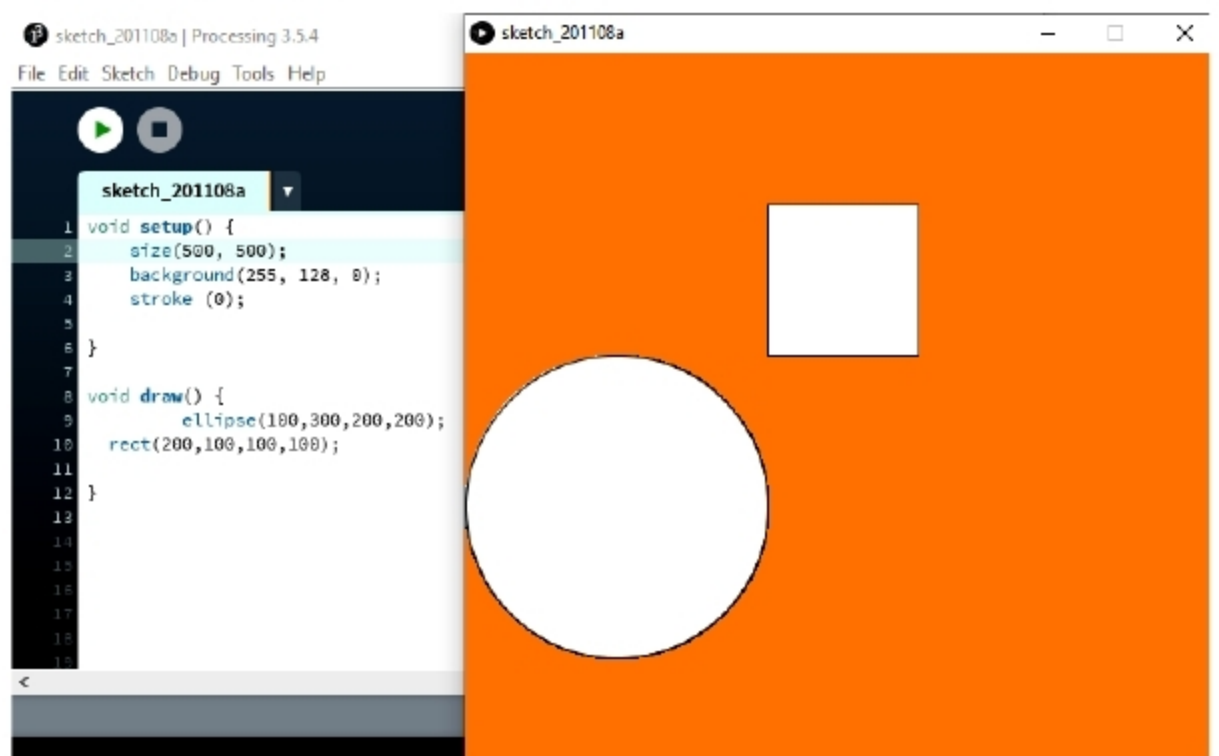


Figure 2: Shapes

```
void draw() {
  ellipse(100, 300, 200, 200);
  rect(200, 100, 100, 100);
}
```

There are dozens of different functions to display geometric shapes in Processing. I will now use two of them: *rect()* and *ellipse()*. This version sets the window size to 500x500 pixels, background to bright orange, and draws a rectangle and ellipse in white by setting the stroke colour to black.

The problem with objects drawn so far is that they are static. How do we make them move?

The answer is simple! It's by using variables to increment/decrement coordinates.

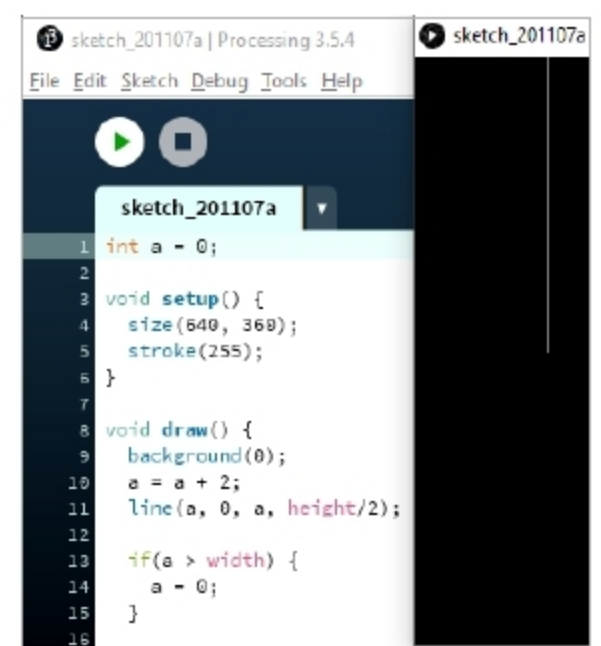


Figure 3: Line animation

Variables are names that hold values. To create a variable, you give it a type, a name, and a value.